Limitations and Open Questions

1. **Requires camera intrinsics at inference.** Unlike UniDepth (CVPR 2024) and Depth Pro (Apple, 2024), which estimate intrinsics internally. Users must provide or estimate f.

2. **Single-frame only.** No temporal consistency for video. Sequential frames can produce flickering depth maps.

3. **Outdoor normal data scarcity persists.** The consistency loss helps but does not fully close the gap; outdoor normal quality lags indoor.

4. **Canonical focal length $f^c = 1000$ is ad hoc.** The paper provides no theoretical justification for this choice. Sensitivity analysis would strengthen the claim.

5. **Newer competitors are closing the gap.**
   - **MoGe** (CVPR 2025): jointly predicts 3D points + normals + FOV
   - **Depth Pro**: metric depth without intrinsics at 0.3s per image